

lecture_12

August 31, 2026

1 Lecture 12: Non-Linear Programming - Principles

{note} Lectures 1-11 built a complete toolkit for **linear** optimization: the objective function and every constraint were straight lines (or hyperplanes). Many real transportation problems - signal timing, congestion pricing, facility siting, vehicle routing - are not linear: costs rise non-proportionally with volume, constraints are products or ratios of decision variables, and the feasible region is no longer a convex polygon. This lecture introduces the landscape of Non-Linear Programming (NLP): what makes a problem non-linear, how non-linearity changes the geometry and the solution strategy, and which families of NLP methods we will study in Lectures 13-22.

Learning Objectives

By the end of this notebook, you will be able to: 1. Recognise the structural differences between linear and non-linear programs, and classify NLP instances by problem type (unconstrained/constrained, convex/non-convex). 2. Identify local and global optima geometrically and analytically, and explain why LP methods fail in non-linear settings. 3. Map real transportation optimization problems to the appropriate NLP framework and solver family covered in this module.

Prerequisites: LP Formulation (Lecture 3), Graphical Solution Method (Lecture 4), Simplex Method (Lecture 7), Duality (Lecture 9)

Estimated time: 90 minutes (including in-class exercises)

1.1 Why Non-Linear Programming?

Every LP we solved in Lectures 3-11 rested on a silent assumption: the relationships between decision variables are proportional and additive. Doubling the number of buses doubles the cost. Sending twice as much freight doubles fuel use. Adding one more worker doubles output. In practice this linearity assumption breaks down in three common ways.

Diminishing Effects: Every additional vehicle on the Sardar Patel Road road reduces traffic efficiency polynomially.

Fixed Assets: Opening a new freight terminal in Chennai incurs a fixed cost regardless of volume.

Interaction Effects. The fuel consumption of a truck on a hilly highway is a function of speed, acceleration, and load.

1.1.1 Example 1

CMRL wants to set green-time splits (x_1, x_2) at a junction to minimize total vehicle delay. Webster's delay formula gives:

$$D_i(x_i) = \frac{C(1 - x_i/C)^2}{2(1 - \lambda_i x_i/C)} + \frac{\lambda_i^2}{2\mu_i(1 - \lambda_i)}$$

where C is cycle length, λ_i arrival rate, μ_i saturation flow.

1.1.2 Example 2

A logistics firm wants to locate a transshipment hub at coordinates (x, y) to minimize total weighted distance to n customer locations (a_k, b_k) with demand weights w_k :

$$\min_{x,y} f(x, y) = \sum_{k=1}^n w_k \sqrt{(x - a_k)^2 + (y - b_k)^2}$$

1.1.3 Example 3

The Beckmann formulation of Wardrop's user equilibrium assigns traffic volumes v_a to roads to solve:

$$\min_v \sum_{a \in \mathcal{A}} \int_0^{v_a} t_a(u) du$$

where $t_a(\cdot)$ is the BPR travel-time function.

Key question: once we leave the land of linear programs, what tools do we need to solve non-linear programs?

1.2 Standard Form

A Non-Linear Program is an optimization problem of the form:

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m \\ & x \in \mathbb{R}^n \end{aligned}$$

where at least one of f or g_i is non-linear in x . Equality constraints $g_j(x) = 0$ are handled exactly as in LP — converted to a pair $g_j(x) \leq 0$ and $-g_j(x) \leq 0$ — so no separate equality row is needed.

Notation:

Symbol	Meaning
$x = (x_1, \dots, x_n)^\top$	Vector of n decision variables

Symbol	Meaning
$f : \mathbb{R}^n \rightarrow \mathbb{R}$	Objective function
$g_i : \mathbb{R}^n \rightarrow \mathbb{R}$	Constraint function (≤ 0 form; equalities handled by conversion)
$\nabla f(x)$	Gradient of f at x — vector of first partial derivatives
$\nabla^2 f(x)$	Hessian of f at x — matrix of second partial derivatives
$\mathcal{F}$	Feasible region: $\{x : g_i(x) \leq 0 \forall i\}$

“{note} The LP is a special case where all functions are linear. In compact matrix–vector notation ($\mathbf{c} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$):

$$\max_{\mathbf{x} \geq \mathbf{0}} \quad \mathbf{c}^\top \mathbf{x} \quad \text{subject to} \quad \mathbf{A}\mathbf{x} \leq \mathbf{b}$$

This is the NLP general form specialised to $f(\mathbf{x}) = -\mathbf{c}^\top \mathbf{x}$ (linear objective, negated for min) and $g_i(\mathbf{x}) = \mathbf{A}_i \mathbf{x} - b_i \leq 0$ (linear constraints, where $\mathbf{A}_i$ is the i -th row of $\mathbf{A}$). The familiar LP optimality conditions (shadow prices, reduced costs, duality) become special cases of the NLP conditions (KKT multipliers, Lagrangian duality) we derive in the later lectures. “

1.3 Taxonomy

1.3.1 Constrained vs Unconstrained Optimization

An optimization problem in the absence of any constraints is an unconstrained optimization problem, rendering $\mathbb{R}^n$ as the entire feasible region:

$$\min_{x \in \mathbb{R}^n} f(x)$$

A necessary first-order condition for x^* to be a local or global optimum is that the gradient vanishes: $\nabla f(x^*) = \mathbf{0}$. A point satisfying this condition is called a stationary point — it may be a minimum, maximum, or saddle point. The second-order sufficient condition resolves the ambiguity:

- Minimum: $\nabla^2 f(x^*) \succ 0$ — the Hessian is positive definite
- Maximum: $\nabla^2 f(x^*) \prec 0$ — the Hessian is negative definite
- Saddle: $\nabla^2 f(x^*)$ is indefinite — it has both positive and negative eigenvalues

On the other hand, if at least one constraint is active, restricting $\mathcal{F} \subsetneq \mathbb{R}^n$, we have a constrained optimization problem:

$$\begin{aligned} & \min_{x \in \mathcal{F}} f(x) \\ & \text{subject to} \quad g_i(x) \leq 0, \quad i = 1, \dots, m \\ & \quad \quad \quad x \in \mathbb{R}^n \end{aligned}$$

Here, stationarity of f alone is not sufficient since the optimum may lie on the boundary of $\mathcal{F}$, where the constraint is active. Optimality is instead characterised by the Lagrangian: $\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x)$, with the Karush–Kuhn–Tucker (KKT) conditions requiring $\nabla_x \mathcal{L}(x^*, \lambda^*) = \mathbf{0}$ along with $\lambda_i g_i(x^*) = 0$ and $\lambda_i \geq 0$.

1.3.2 Convex vs Non-Convex Program

In a convex program:

- the feasible set $\mathcal{F}$ is convex — the line segment joining any two of its points lies entirely within it:

$$x_1, x_2 \in \mathcal{F}, \lambda \in [0, 1] \implies \lambda x_1 + (1 - \lambda)x_2 \in \mathcal{F}$$

- the objective f is convex — the chord joining any two points on its graph lies on or above the graph:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2) \quad \forall x_1, x_2, \lambda \in [0, 1]$$

$$\implies f(x_2) \geq f(x_1) + \nabla f(x_1)^\top (x_2 - x_1), \quad \forall x_1, x_2 \in \mathcal{F}$$

$$\implies \nabla^2 f(x) \succeq 0 \quad \forall x \in \mathcal{F}$$

Crucially, every local minimum of a convex program is also a global minimum. Hence, gradient-based methods — which converge to a point where $\nabla f = \mathbf{0}$ — are sufficient to find the globally optimal solution.

If f or $\mathcal{F}$ (or both) are non-convex, we get non-convex program wherein a local minimum need not be a global minimum.

“{note} A point x^* is a local minimum of f on $\mathcal{F}$ if there exists $\epsilon > 0$ such that:

$$f(x^*) \leq f(x) \quad \forall x \in \mathcal{F} \text{ with } \|x - x^*\| < \epsilon$$

A point x^* is the global minimum of f on $\mathcal{F}$ if:

$$f(x^*) \leq f(x) \quad \forall x \in \mathcal{F}$$

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1.4 Why LP Methods Fail for NLP

The Simplex method's power comes from three facts that are unique to LP:

1. Convex Polytope Feasible Region: Every corner (vertex) is a candidate optimal solution, and the Simplex method hops between adjacent vertices along improving edges.
2. Linear Objective: The gradient $\nabla f = \mathbf{c}$ is constant — it never changes direction, so the method always knows which way is downhill.
3. Global Optimum: LP's strong duality theorem guarantees that when Simplex stops (all reduced costs ≥ 0), the current vertex is globally optimal — no other point anywhere in $\mathcal{F}$ can do better.

None of these three properties holds for a general NLP, wherein $\mathcal{F}$ can be non-convex with $\nabla f(x)$ that changes with x rendering potential local optima or saddle points.

1.5 In-Class Exercises

1.5.1 Exercise 1

BMTC is planning bus frequencies x_1, x_2 (buses/hour) on two Bengaluru corridors. Passenger demand on each corridor increases with service frequency at a diminishing rate, giving total revenue $R(x_1, x_2) = 800\sqrt{x_1} + 600\sqrt{x_2}$. BMTC has a fleet availability constraint limiting total deployment to 20 buses/hour, and a minimum service constraint requiring at least 3 buses/hour on each corridor. The operating cost is 1,200 per bus-hour.

1. Write the profit-maximization NLP in standard form (convert to minimization).
2. Classify the problem — constrained or unconstrained, linear or non-linear, convex or non-convex.

1.5.2 Exercise 2

A Delhivery cross-dock facility in Guwahati processes freight arriving at rate λ (tonnes/hour). The variable processing cost increases with throughput: $C(\lambda) = 50\lambda + 2\lambda^2$. The facility also incurs a fixed operating cost of 8,000 per hour and earns revenue of 200 per tonne processed. To satisfy its service-level agreement, throughput must be at least 30 tonnes/hour; equipment capacity limits throughput to 80 tonnes/hour.

1. Write the profit-maximization NLP in standard form (convert to minimization).
 2. Classify the problem — constrained or unconstrained, linear or non-linear, convex or non-convex.
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1.6 Take-Away Exercises

1.6.1 Exercise 1

A NHAI corridor study estimates travel time on Chandigarh-Ambala link using the BPR function:

$$t(v) = 15 \left[1 + 0.15 \left(\frac{v}{1,200} \right)^4 \right] \quad (\text{minutes})$$

where v is traffic volume (vehicles/hour) and capacity is $c = 1,200$ veh/hr.

1. Compute $t(600)$, $t(1,200)$, and $t(2,400)$. Comment on how travel time responds to volume near and above capacity.
2. Differentiate $t(v)$ with respect to v and evaluate $t'(1,200)$. Interpret the result: what does this derivative tell NHAH about marginal congestion at capacity?
3. Is $t(v)$ a convex function of v for $v \geq 0$? Justify by computing $t''(v)$ and checking its sign.

1.6.2 Exercise 2

The traffic volume v on a Mumbai expressway link is assigned to minimize the Beckmann objective:

$$B(v) = \int_0^v t_0 \left[1 + 0.15 \left(\frac{u}{c} \right)^4 \right] du$$

with $t_0 = 20$ min, $c = 2,000$ veh/hr, and the constraint $0 \leq v \leq 3,000$.

1. Is $B(v)$ convex on $[0, 3000]$? Show using $B''(v)$.
2. The demand constraint requires $v \geq 1,500$. Write the constrained NLP in standard form ($g(v) \leq 0$).

1.7 Further Reading

- Bazaraa, M.S., Sherali, H.D., and Shetty, C.M. (2006). *Nonlinear Programming: Theory and Algorithms* (3rd ed.). Wiley — Chapters 1–2 (problem classes, optimality conditions overview).
- Nocedal, J. and Wright, S.J. (2006). *Numerical Optimization* (2nd ed.). Springer — Chapter 1 (introduction), Chapter 2 (fundamentals of unconstrained optimization).
- Wardrop, J.G. (1952). Some theoretical aspects of road traffic research. *Proceedings of the Institute of Civil Engineers*, 1(3), 325–362. — Origin of the traffic equilibrium conditions.
- BPR (1964). *Traffic Assignment Manual*. US Bureau of Public Roads, Washington DC. — Source of the BPR travel-time function.
- SciPy documentation: `scipy.optimize.minimize` — docs.scipy.org